

CHRISTINA DURÓN

✉ christina.duron@pepperdine.edu

☎ (310) 506 – 4832

🌐 <https://cduron.info>

ACADEMIC EMPLOYMENT

Associate Professor of Mathematics <i>Natural Science Division of Seaver College, Pepperdine University</i>	Aug 2026 – Present
Assistant Professor of Mathematics <i>Natural Science Division of Seaver College, Pepperdine University</i>	Aug 2022 – July 2026
Postdoctoral Research Associate <i>Department of Mathematics, University of Arizona</i>	Aug 2019 – May 2022
High School Teacher <i>Mathematics Department, The Webb Schools of California</i>	Aug 2013 – June 2019

RESEARCH INTERESTS

Network Theory; Network Dynamics; Statistical Analysis and Modeling of Complex Networks; Mathematical-Biology

EDUCATION

Claremont Graduate University <i>Ph.D. in Mathematics</i> • Thesis: The Distribution of Betweenness Centrality in Exponential Random Graph Models • Advisors: Dr. Ami Radunskaya (Professor, Pomona College) and Dr. Johanna Hardin (Professor, Pomona College)	May 2019
University of Washington <i>Master in Science in Applied Mathematics</i>	June 2013
Swarthmore College <i>Bachelor of Arts in Mathematics; Minor in Computer Science</i>	May 2012

PUBLICATIONS *

13. Kravitz H, **Durón C**, Nieves B, Brio M. (2025). Data-driven optimization and parameter estimation for a metric graph epidemic model with applications to COVID-19 spread in Poland: A real-world example of optimization for a challenging Rosenbrock-type objective function. *An International Journal of Optimization and Control: Theories & Applications (IJOCTA)*, 15(4). doi: [10.36922/IJOCTA025220106](https://doi.org/10.36922/IJOCTA025220106)
12. **Durón C**, Kravitz H, Brio M. (2025). Edge percolation centrality: A new measure to quantify the influence of edges during percolation in networks. *PLoS One* 20(9): e0331475. doi: [10.1371/journal.pone.0331475](https://doi.org/10.1371/journal.pone.0331475)
11. **Durón C**, Kravitz H, Brio M. (2025). Kolmogorov–Smirnov-Based Edge Centrality Measure for Metric Graphs. *Dynamics*, 5(2), 16. doi: [10.3390/dynamics5020016](https://doi.org/10.3390/dynamics5020016)
10. Pfeffer D, **Durón C**. (2025). An Unlikely Duo: Injecting Art Projects in the Mathematics Classroom. *Journal of Humanistic Mathematics*, 15(1), 87-108. doi: [10.5642/jhummath.RDDV3141](https://doi.org/10.5642/jhummath.RDDV3141)

* Authors are ordered by contribution.

† Undergraduate student.

9. Kravitz H, **Durón C**, Brio M. (2024). A Coupled Spatial-Network Model: A Mathematical Framework for Applications in Epidemiology. *Bulletin of Mathematical Biology*, 86(132): 1 – 35. doi: [10.1007/s11538-024-01364-3](https://doi.org/10.1007/s11538-024-01364-3)
8. **Durón C**, Swansen B[†], Kravitz H. (2024). Difference Approximation. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat08440](https://doi.org/10.1002/9781118445112.stat08440)
7. Sullivant N, **Durón C**, Pfeffer D. (2023). From Mirrors to Wallpapers: A Virtual Math Circle Module on Symmetry. *Journal of Math Circles*: Vol. 3: Iss. 1, Article 1. Available at <https://digitalcommons.cwu.edu/mathcirclesjournal/vol3/iss1/1>.
6. **Durón C**. (2022). Adaptive Quadrature. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat08388](https://doi.org/10.1002/9781118445112.stat08388)
5. **Durón C**, Farrell A. (2022). A Mean-Field Approximation of SIR Epidemics on an Erdős-Rényi Network Model. *Bulletin of Mathematical Biology*, 84(7): 1 – 19. doi: [10.1007/s11538-022-01026-2](https://doi.org/10.1007/s11538-022-01026-2)
4. **Durón C**. (2021). Linear Algebra, Computational. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat00459.pub2](https://doi.org/10.1002/9781118445112.stat00459.pub2)
3. **Durón C**. (2020). Heatmap Centrality: A New Measure to Identify Super-Spreader Nodes in Scale-Free Networks. *PLoS ONE*, 15(7): e0235690. doi: [10.1371/journal.pone.0235690](https://doi.org/10.1371/journal.pone.0235690)
2. **Durón C**, Pan Y, Gutmann D.H., Hardin J, & Radunskaya A. (2018). Variability of Betweenness Centrality and Its Effect on Identifying Essential Genes. *Bulletin of Mathematical Biology*, 81(9): 3655 – 3673. doi: [10.1007/s11538-018-0526-z](https://doi.org/10.1007/s11538-018-0526-z)
1. Pan Y, **Durón C**, Bush E.C., et al. (2018). Graph Complexity Analysis Identifies an ETV5 Tumor-Specific Network in Human and Murine Low-Grade Glioma. *PLoS ONE*, 13(5): e0190001. doi: [10.1371/journal.pone.0190001](https://doi.org/10.1371/journal.pone.0190001)

UNDER REVIEW *

1. Kravitz H, **Durón C**, Brio M. Epidemic dynamics under pulse population exchange. *Dynamics*.

IN PREPARATION *

3. **Durón C**, Kravitz H, Brio M, Parkinson C, Lucas B[†]. Epidemic Thresholds on Adaptive Networks with Noncompliance. *Mathematical Biosciences*.
2. O'Brien E, **Durón C**. The Wasserstein metric as a tool to assess burn-in and convergence. *Statistical Science*.
1. **Durón C**, Mello J[†], Kamps K[†]. Network Evaluation of Influential Sensors on Critical Infrastructures. *The Journal of Dam Safety*.

RESEARCH POSITIONS

Graduate Research Assistant

Pomona College

Jan 2017 – June 2018

- NIH (1R01-CA195692-01) funding under Dr. Ami Radunskaya and Dr. Johanna Hardin

Jet Propulsion Laboratory (JPL) Intern

California Institute of Technology

June 2015 – August 2015

- Implemented the Extended Kalman Filter (EFK) and incorporated inter-robot measurements to improve the state estimation and localization of autonomous vehicles

Mathematical and Theoretical Biology Institute Researcher

Arizona State University

June 2011 – July 2011

- Developed a mathematical model for the evaluation and analysis of the air pollution in Los Angeles

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† Undergraduate student.

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TEACHING EXPERIENCE

Instructor of Record

Pepperdine University

- **MATH 450: Mathematical Statistics** *Spring 2024*
- **MATH 350: Mathematical Probability** *Fall 2025, Fall 2023*
- **MATH 345: Numerical Methods** *Spring 2026*
- **MATH 299: Directed Studies** *Summer 2026*
- **MATH 260: Linear Algebra** *Fall 2026, Spring 2026, Fall 2024, Spring 2023*
- **MATH 150: Calculus I** *Fall 2026, Fall 2025, Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2022*

Instructor of Record

University of Arizona

- **MATH 491: Undergraduate Teaching Assistantship (UTA) Seminar** *Fall 2021 – Spring 2022*
- **MATH 196M: Calculus I Supplementary Seminar** *Spring 2022*
- **MATH 396L: Wildcats Proofs Workshop** *Spring 2022*
- **MATH 464: Theory of Probability** *Fall 2021*
- **MATH 363: Introduction to Statistical Methods** *Spring 2021*
- **MATH 129: Calculus II** *Fall 2020*
- **MATH 475A: Mathematical Principles of Numerical Analysis** *Fall 2020*
- **MATH 163: Basic Statistics** *Spring 2020*
- **MATH 122B: First Semester Calculus** *Fall 2019*
- **MATH 196L: Precalculus Supplementary Seminar** *Fall 2019*

Instructor of Record

The Webb Schools of California

- **Advanced Placement Computer Science Principles** *Fall 2018 – Spring 2019*
- **Honors Precalculus** *Fall 2014 – Spring 2019*
- **Introduction to Computer Programming with Python** *Fall 2014 – Spring 2018*
- **Precalculus** *Fall 2013 – Spring 2019*
- **Integrated Mathematics 2** *Fall 2013 – Spring 2014*

PRESENTATIONS

Student-Directed

4. **Modeling Epidemics with Noncompliance: A Discrete-Time Mean-Field Approach** April 2026
Pepperdine University; Seaver College Research and Scholarly Research Symposium
3. **Network Evaluation of Influential Sensors: A Proposed Approach** March 2024
Pepperdine University; Seaver College Research and Scholarly Research Symposium
2. **Using Networks to Evaluate an Infrastructure's Condition During Seismic Activity: A Proposed Approach** Jan 2024
Joint Mathematics Meeting: AMS - PME Undergraduate Student Poster Session
1. **Using Networks to Evaluate an Infrastructure's Condition During Seismic Activity: A Proposed Approach** July 2023
Pepperdine University, Natural Science Division Summer Research

Contributed

9. **Edge Percolation Centrality: Measuring Dynamic Influence on Networks** March 2026
American Mathematical Society Special Session on Metric Graphs: Theory and Applications
8. **Digitally-Enabled, Evidence Based Homework: Developing a Custom Platform to Improve Calculus Learning** January 2026
Joint Mathematics Meeting Special Session on Digitally-Enabled Evidence-Based Teaching Practices
7. **A Coupled Spatial-Network Model for Epidemiology** July 2024
Frankfurt School of Finance and Management Workshop on Metric Networks
6. **Math Circles: Elevating Student Engagement and Mathematical Enjoyment Outside the Classroom** Jan 2024
University of Arizona Mathematics Educator Appreciation Day Conference (MEAD)
5. **Incorporating Undergraduate Students in Research on Assessing Dam Safety Using Network Analysis** Jan 2024
Joint Mathematics Meeting Special Session on Navigating the Benefits and Challenges of Mentoring Students in Data-Driven Undergraduate Research Projects
4. **Tiling a Chessboard: A Problem Adapted for the Virtual Math Circle** Jan 2023
Joint Mathematics Meeting Special Session on Math Circle Activities as a Gateway into Mathematics
3. **A Mean Field Approximation of SIR Epidemics on an Erdős-Rényi Network Model** May 2021
Los Alamos-Arizona Days Conference (Virtual Poster)
2. **Identifying Super-Spreader Nodes in Scale-Free Networks using Network Centrality Measures** Sept 2020
Arizona Postdoctoral Research Conference (Virtual Talk)
1. **Identifying Treatment Targets for Pediatric Gliomas using Network Centrality Measures** June 2020
SIAM Conference on the Life Sciences (Virtual Talk)

Seminar

4. **A Coupled Spatial-Network Model for Epidemiology** Nov 2023
University of Florida, Health Systems Medicine Seminar (Virtual Talk)
3. **Network Centrality: Theory to Applications** Oct 2021
Arizona State University, Mathematical Biology Seminar (Virtual Talk)
2. **Heatmap Centrality: A New Measure to Identify Super-Spreader Nodes in Scale-Free Networks** Feb 2021
Claremont Colleges and University of Utah, Joint Applied Mathematics Seminar (Virtual Talk)
1. **Network Data Analysis Techniques on DESeq and RNASeq Data** Nov 2019
University of Arizona, TRIPODS Research Working Group 6 - Analyzing large-scale point-set data

Industry

1. **Network Evaluation of Influential Sensors: A Proposed Approach[‡]** March 2024
California Department of Water Resources
2. **Network Evaluation of Influential Sensors: A Proposed Approach[‡]** Nov 2023
Southern California Edison, Dam & Public Safety Group

PROFESSIONAL DEVELOPMENT

Program

- **American Institute of Mathematics SQuaRE Program** July 2026, April 2025
California Institute of Technology
 - Member of a multi-year research collaboration focused on creating computational frameworks to predict epidemic transitions and optimize social measures for disease mitigation.
- **Project NExT (Green 2023)** Aug 2023 – Aug 2024
Mathematical Association of America (MAA)
 - Focused on all aspects of an academic career: improving the teaching and learning of mathematics, engaging in research and scholarship, finding exciting and interesting service opportunities, and participating in professional activities

Certification

- **Diversity, Equity, and Inclusion in the Workplace** May 2021
University of South Florida
 - Focused on ways that organizations can create a more diverse workplace, address equity issues, and foster inclusivity
- **Effective Online Discussions** June 2020
University of Arizona
 - Developed strategies for designing and facilitating effective online discussions that deepen learning, expand student exposure to curriculum, and increase student engagement
- **Teaching the Large Online Course** June 2020
University of Arizona
 - Developed instructional practices for encouraging student engagement and motivation in a large online class, as well as for effectively managing administrative tasks such as monitoring student progress and conducting assessments

Workshops

- **Workshop on Metric Networks** July 2024
Frankfurt School of Finance and Management
- **Broadening Participation: 2024 Mathematical and Physical Sciences Workshop for New Investigators** July 2024
National Science Foundation
- **Network Modeling for Epidemics** Aug 2020
University of Washington
- **BioBridge Clinic** Jan 2020
University of California, Irvine
- **Computational Genomics Summer Institute** May 2019
University of California, Los Angeles

[‡] Presented with undergraduate students.

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SERVICE

Mentoring and Advising

- **Undergraduate Research Advisor** (Team of 1 undergraduate)
Pepperdine University Summer 2026 – Present
- **Undergraduate Research Advisor** (Team of 3 undergraduates)
Pepperdine University Fall 2025 – Spring 2026
- **Tidy Thursday Advisor**
Pepperdine University Spring 2026, Fall 2024
- **Undergraduate Research Advisor** (Team of 2 undergraduates)
Pepperdine University Summer 2023 – Fall 2024
- **Math Club Vice-Principal Faculty Advisor**
Pepperdine University Spring 2023 – Fall 2023
- **Undergraduate Student Mentor**
Pepperdine University, Faculty/Student Mentor Program Fall 2025, Spring 2023 – Fall 2023
- **Undergraduate Student Mentor**
University of Arizona, Women in Science and Engineering (WISE) Program Fall 2021 – Spring 2022
- **Graduate Student Mentor**
University of Arizona, Association for Women in Mathematics (AWM) Mentor Network Fall 2021 – Spring 2022
- **Undergraduate Research Advisor**
University of Arizona Fall 2020 – Spring 2021
- **Mathematics Undergraduate Teaching Assistantship (UTA) Program Mentor**
University of Arizona Fall 2020 – Spring 2021
- **Mathematical Modeling Group Mentor** (Team of 4 undergraduates)
University of Arizona Spring 2020
- **Math Club Advisor**
The Webb Schools of California Fall 2017 – Spring 2019

University and Departmental Service

- **Creator and principal faculty advisor for Tidy Thursday**
Pepperdine University Spring 2026, Fall 2024
 - An unofficial student club focused on exploring and analyzing datasets in **R** to improve data analysis and visualization skills.
- **Institutional Review Board, Natural Science Representative**
Pepperdine University Fall 2023 – Present
 - Review research study protocol/application materials (exempt, expedited, and full review) and evaluate them from the perspective of the regulatory criteria for approval.
- **Admissions and Scholarship Committee**
Pepperdine University Fall 2025 – Present, Fall 2023 – Fall 2024
 - Review student applications and select Faculty-Staff Scholars based on merit, including academic performance and recommendations.
- **Co-Instructor of RISE/Connection/Welcome Chapel**
Pepperdine University Fall 2025 – Present, Fall 2023 – Fall 2024
 - Facilitated discussions on different aspects of science from the Christian perspective, primarily in the disciplines of history, physics, astronomy, and mathematics, with students over five sessions

- **Member of Hiring Committee, Assistant Professor of Mathematics (AY 24/25)**
Pepperdine University

 - Served on the seven-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom and in-person interviews on campus, collecting and consolidating information about the interviews, and making final recommendations

Fall 2023 – Spring 2024
- **Member of Hiring Committee, Visiting Assistant Professor of Mathematics (AY 23/24)**
Pepperdine University

 - Served on the five-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom, collecting and consolidating information about the interviews, and making final recommendations

Summer 2023
- **Member of Hiring Committee, Assistant Professor of Mathematics (AY 23/24)**
Pepperdine University

 - Served on the seven-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom and in-person interviews on campus, collecting and consolidating information about the interviews, and making final recommendations

Fall 2022 – Spring 2023
- **Member of Review Committee, Excellence in Postdoctoral Mentoring Award**
University of Arizona

 - Served on the three-member review committee that determined the recipient of the 2022 Excellence in Postdoctoral Mentoring Award

Spring 2022
- **President, Postdoctoral Governance**
University of Arizona

 - Serve as an in-between for the postdocs and the Postdoctoral Committee, and organize the postdoctoral professional development seminar topics and panels

Fall 2021 – Spring 2022
- **Mathematics Undergraduate Teaching Assistantship (UTA) Program, Director**
University of Arizona

 - Coordinate the mentorship of the UTA's, and run the weekly professional development seminar

Fall 2021 – Spring 2022
- **Mathematics Undergraduate Teaching Assistantship (UTA) Program, Co-Director**
University of Arizona

 - Supported the Director of the UTA Program, and was responsible for additional duties related to the weekly professional development seminar

Fall 2020 – Spring 2021
- **Vice President, Postdoctoral Governance**
University of Arizona

 - Supported the President of the Postdoctoral Governance, and was responsible for additional duties related to the postdoctoral professional development seminars

Fall 2020 – Spring 2021
- **Non-Academic Liaison, Postdoctoral Governance**
University of Arizona

 - Organized a panel pertaining to non-academic careers for the postdoctoral professional development seminar

Spring 2020

Service to the Discipline

- **American Mathematical Society Spring Western Sectional Meeting 2026, Co-Organizer of Special Session**
Boise State University

 - “Metric Graphs: Theory and Applications”

March 2026
- **Webinar Panelist**
Transform Learning

 - “Thinking About Thinking: Using Formative Practice to Grow Metacognitive Learners”

Feb 2026

- **Joint Mathematics Meeting 2024, Co-Organizer of MAA Project NExT Session** *Jan 2024*
Pepperdine University
 - “Making Student Thinking Visible with Team-Based Inquiry Learning”
- **MAA MathFest 2022, Co-Organizer of Themed Contributed Paper Session** *Aug 2022*
University of Arizona
 - “Math Circles: Talks about Mathematical Joy, Inspirations, and Data-Driven Lessons Learned”
- **Reviewer for:**
 - Journal of Statistics and Data Science Education *Feb 2026*
 - Applied Network Science *Dec 2025*
 - Complexity *March 2023*
 - Indian Journal of Discrete Mathematics *Nov 2020*
 - DNA and Cell Biology *Jan 2020*
 - Revista de Matemática: Teoría y Aplicaciones *Oct 2019*
- **Mathematics and MATLAB Summer Workshop, Co-Coordinator** *June 2016 – June 2018*
Claremont Graduate University
- **Mathematics and MATLAB Summer Workshop, Co-Instructor** *June 2016 – June 2017*
Claremont Graduate University

Outreach

- **How I Found My Network: My Path to Mathematics** *Nov 2021*
Arizona State University
 - Keynote address for Sonia Kovalevsky Day
- **Tucson Math Circle** *Aug 2019 – Present*
University of Arizona
 - Co-develop materials and co-run the university sponsored weekly program designed to get middle school students excited about mathematics through hands-on exploration and discovery
- **Association for Women in Mathematics (AWM): Sonia Kovalevsky Day** *April 2021*
University of Arizona
 - Developed materials and co-ran a workshop designed to bolster female high school and middle school students’ passion and enthusiasm for mathematics in a supportive environment
- **The Seven Bridges of Königsberg** *Nov 2022*
Pepperdine University
 - Talk given to Math Club during Tuesday Tea
- **Using Network Centrality Measures to Identify Unknown Regulatory Pathways in Pediatric Glioma** *Sept 2020*
University of Arizona
 - Talk given to The MathCats Club (undergraduate math club)

HONORS AND AWARDS

The Teaching and Service Award <i>University of Arizona, Department of Mathematics</i>	<i>April 2022</i>
The Jean E. Miller Excellence in Teaching Award <i>The Webb Schools of California</i>	<i>June 2018</i>
The Thompson and Vivian Webb Excellence in Teaching Award <i>The Webb Schools of California</i>	<i>June 2015</i>
The Heinrich W. Brinkmann Mathematics Prize <i>Swarthmore College</i>	<i>June 2012</i>

FUNDING

Research Grants

- **Academic Year Undergraduate Research Initiative** (\$1,000) *Sept 2025, Sept 2023*
Pepperdine University
- **Collaborative Research Grant for Postdocs** (\$1,500) *June 2020*
University of Arizona

Travel Awards

- **AIM SQuaRE Research Collaboration** *July 2026, April 2025*
American Institute of Mathematics (AIM)
- **TDA-BIO** (\$1,000) *Oct 2016*
ACM Conference on Bioinformatics, Computational Biology, and Health Informatics

Fellowships

- **Clinic Fellowship** (\$900) *Jan 2020*
University of California, Irvine
- **Daniel Pick Fellowship** (\$10,000) *Oct 2017*
Claremont Graduate University
- **Joseph and Elizabeth Peeler Endowed Fellowship** (\$32,570) *Aug 2015 – June 2017*
Claremont Graduate University
- **CGU Mathematics Fellowship** (\$13,700) *Aug 2014 – June 2015, June 2017*
Claremont Graduate University
- **CGU Minority Fellowship** (\$2,000) *Aug 2014 – June 2016*
Claremont Graduate University

SKILLS

Programming Languages

- C (*Moderate proficiency*)
- C++ (*Moderate proficiency*)
- MATLAB (*Proficient*)
- Python (*Proficient*)
- R (*Proficient*)

Scientific Applications

- GitHub
- LaTeX
- RSweave